Technological Development of Next-Generation 2.5D/2.xD Packages, Panel-Level Interposers, and Related Materials through JOINT2 Activities

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Abstract

Resonac has established the semiconductor advanced packages evaluation platform "JOINT2" to promote technological development for next-generation 2.5D/2.xD packages and interposer. In this study, we evaluated the issues and the impact of various material properties on 1) fine bump interconnection, 2) fine wiring formation, and 3) high-reliability large-scale substrates through the prototyping of interposers and large packages. Based on these findings, we present the latest trends and prospects in material development that supports advanced packaging technology.

Author Keywords

advanced package; packaging; interposer; 2.5D; 2.xD; 2.3D; interconnection; micro bump; flip chip, underfill; redistribution layer; fine wiring; glass carrier; chip embedded; large body; large substrate; electrical reliability; crack resistance;

1. Introduction

In 2018, Resonac established the Packaging Solution Center as a new research and development hub aimed at providing one-stop solutions for customers. In October 2021, the company launched JOINT2, a co-creation-based package evaluation platform in collaboration with leading industry partners, to accelerate the development of advanced materials, equipment, and substrates for 2.xD and 3D packages. In recent years, 2.xD and 3D packaging technologies have required increasingly dense interconnections between chips and components, driving the need for further miniaturization of wiring pitch and vertical interconnect dimensions[1]. At the same time, to achieve higher performance, the level of chip integration has increased, resulting in a trend toward larger package sizes. To address these demands, Resonac has been developing fine-pitch vertical and horizontal interconnect technologies, as well as conducting research on the manufacturing processes and reliability of ultra-large 2.5D advanced packages. In the area of interconnect technology, our efforts have focused on micro-bump formation and bonding processes, fluxless bonding techniques, and encapsulation materials capable of filling narrow gaps[2-4]. In the following sections, we present a comprehensive overview of the JOINT2 project, including its objectives, technical approaches, and recent achievements.

2. Project introduction

"JOINT2" co-creation platform for advanced package evaluation, led by Resonac, 14 major material, equipment, and substrate companies are collaborating to promote the technological development of 2.5D/2.xD packages. Fig.1 shows a list of companies participating in JOINT2. JOINT2 is active in three working groups. WG-A is verifying fine bump interconnection and evaluating the limits of micro-bumps with a target bump pitch of 10 microns. WG-B aims to demonstrate 1 micron level line and space in the formation of fine wiring on organic interposers. WG-C is examining large, high-reliability packages, aiming to expand to 140 mm square. The concrete results of JOINT2 so far include the establishment of a 10μm-pitch fine bump connection in WG-A using a test vehicle. We designed and fabricated these interposers and demonstrated underfill filling performance. In WG-B, we prototyped multilayer RDL interposers and die-embedded interposers using a panel process. In WG-C, we prototyped a 120 x 120 mm 2.5D large-scale package substrate and achieved 1,000 cycles in a motherboard-level reliability evaluation. Through these co-creation activities, we are accelerating the development of next-generation packaging technologies. This paper summarizes JOINT2's key technical achievements, and discusses remaining challenges and directions for future work.

Fig. 1 List of "JOINT2" participating companies

3. Experiment and equipment

The processes were carried out under the following equipment and conditions: flip-chip thermo-compression bonding of dies to a silicon interposer with electroplated Cu/Ni/SnAg microbumps; capillary underfill dispensing with the stage maintained at 100°C; EMC compression molding (WCM330, Yamaha Robotics); post-mold surface grinding (DAG810, DISCO) and temporary bonding of a TBA-coated glass carrier using a wafer bonder (AIMECHATEC). With carrier support, the interposer was backside-thinned by grinding to the target thickness, followed by dielectric formation. Panel-level RDL was fabricated via SAP: coating a photo-imageable dielectric (PID) with a panel-capable coater (viscosity controlled) and curing; Ti/Cu seed deposition (25 nm/150 nm); resist patterning considering seed side-etch; Cu electroplating, resist stripping, and seed etching. Fine Cu lines with L/S = 1.5/1.5 μm and three RDL layers were achieved without delamination. Cu/SnAg bumps were formed by SAP in the same module. Debonding from the glass carrier was performed with a laser debonding tool (AIMECHATEC), and final singulation used a dicing saw (DAD3360, DISCO). Key process parameters (temperature, time, thickness, pitch) were controlled to ensure repeatability and stability. For panel-level die-embedded interposers, a laser-releasable temporary layer was

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formed on a glass carrier, followed by coating a PID. Cu pillars were electroformed on the dielectric. Dies were mounted using a panel-level flip-chip bonder with die-attach film (DAF), and DAF was cured in a pressure oven. Panel-size compression molding and planarization of the molding compound, Si, and Cu pillars were followed by dielectric formation. Ti/Cu seed layers (25 nm/150 nm) were deposited, and Cu wirings with L/S = 5/5–10/10 μm were fabricated by SAP. Debonding and singulation were performed as above. Electrical reliability was evaluated on fine HBM–logic Cu lines (L/S = 1.5/1.5 μm) with and without an inorganic protection layer using comb structures (≥150 lines per anode/cathode). Biased-HAST was conducted at 130°C, 85% RH, and 3.3 V with continuous resistance monitoring; failure was defined as <1 × 10^6 Ω. Temperature cycling per JESD22-A104F (−55°C to 125°C, 15 min dwells) was performed on 120 × 120 mm² 2.5D packages mounted on motherboards to assess the effects of CTE mismatch and crack resistance due to structural/material selections.

4. Fine bump interconnection (WG-A)

In 2.xD packages designed to enable high-speed inter-chip communication, the demand for high I/O counts is increasing, driving further miniaturization of interconnections. WG-A has targeted a minimum bump pitch of 10 μm and conducted bonding trials at 20 μm and 15 μm pitches, comparing two molding approaches: (1) conventional capillary underfill (CUF) injection combined with solid epoxy molding compound (EMC) molding, and (2) liquid compression mold underfill (LCMUF) molding. In both processes, setting the average filler particle size to below 1 μm enabled successful filling of 15 μm-pitch bumps, and ultimately made 10 μm-pitch filling possible. Fig.1 shows an example of filler particle size and filling defects. This suggests that as gaps become narrower in the future, it will be necessary to further reduce the filler particle size. While CUF relies on capillary action and is therefore more difficult to control, LCMUF allows pressure-assisted filling using a molding machine. Generally, reducing filler particle size and content improves flowability for fine-pitch bump gap filling, but this also increases the coefficient of thermal expansion (CTE) and shrinkage, potentially worsening wafer warpage. Thus, achieving an optimal balance between filling performance and wafer warpage is critical for fine-pitch bumps. In this study, we evaluated the filling capability and warpage performance of LCMUF molding and CUF + EMC molding for fine-pitch bumps, and investigated optimal process conditions. Fig.2 shows the results of 20-10 μm pitch filling for the CUF process. Through these WG-A activities, we demonstrated that addressing interconnection pitch miniaturization requires not only modifying material properties, but also validating processes at the wafer or interposer level, thereby highlighting the effectiveness of the co-creation platform in supplying optimized materials.

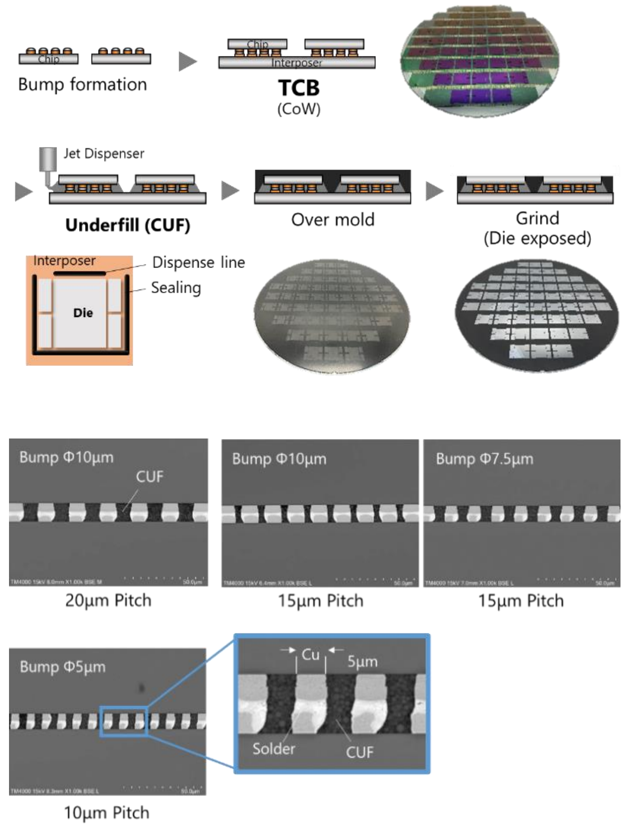


Fig. 2 Fine bump interconnection using the CUF process

5. Fine circuitry interconnection (WG-B)

WG-B conducted the fabrication of two types of interposers with the aim of verifying fine wiring formation technology. The first type was a redistribution layer (RDL) interposer utilizing multilayer Cu wiring, fabricated on a panel with dimensions of 320 mm × 320 mm. The second type was a chip-embedded interposer, fabricated on a panel with dimensions of 515 mm × 510 mm. Fig. 1 illustrates the fabrication of the RDL interposer, which consists of three layers of redistribution wiring. In the fabrication process, a PID was applied onto a glass carrier using a spin coating method, followed by the formation of opening patterns. Cu wiring was then formed using the semi-additive process (SAP). The appearance of the fabricated panels, SEM images of the fine wiring sections, and the yield results for each layer are summarized in Table 1. Under optimized process conditions, a minimum wiring pitch of 3 μm was achieved.

LCMUF Properties

Item	LCMUF-A	LCMUF-B
Filler Content	77.0wt%	83.5wt%
Filler size Average	0.3 μm	1.7 μm
Tg	129°C	154°C
CTE (<Tg)	23ppm/°C	13ppm/°C
Modulus (<Tg)	11 GPa	14 GPa

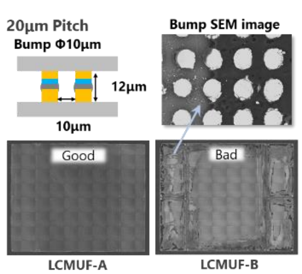


Fig. 1 Effect of filler size on filling properties

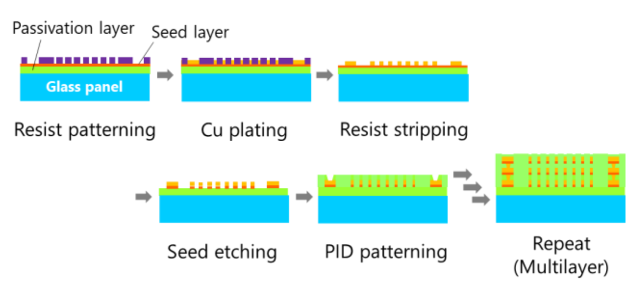


Fig.3 Panel level semi-additive multi-layer wiring process

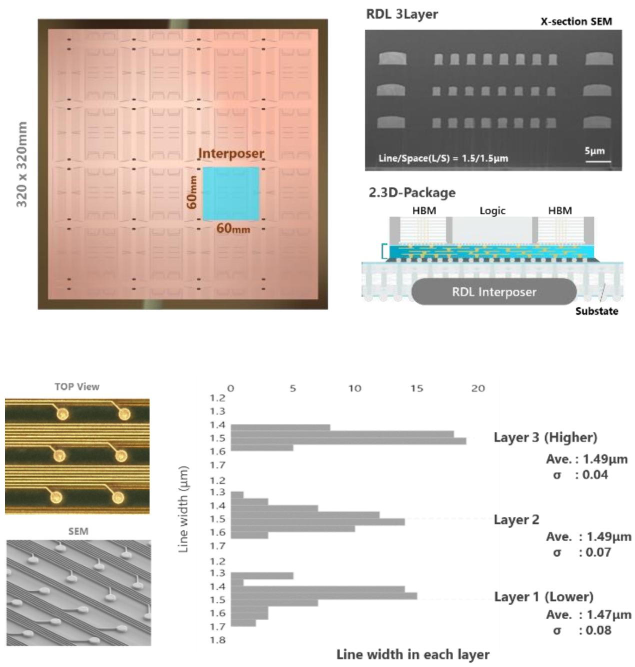


Fig.4 Appearance of panel level multi-layer interposer

To achieve further miniaturization, we considered the damascene process in addition to SAP. In the damascene process, we used a positive type PID. As shown in Fig.5, a trench pattern was formed using the PID damascene method, followed by a seed layer formed by sputtering, and a wiring pattern formed by copper plating, after which CMP was performed. Fig.5 shows cross-sectional photographs of 2 μm pitch wiring patterns after surface polishing by CMP. By using the trench process, we were able to achieve 1 μm-level wiring using a PID damascene process.

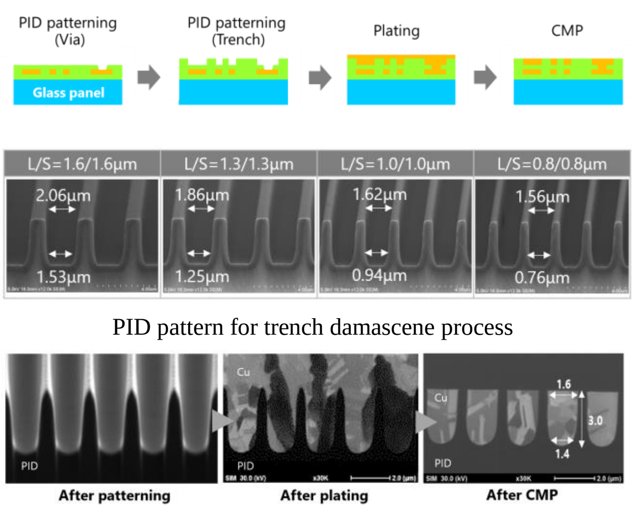


Fig.5 Cross section image of trench process (2μm pitch)

Next, we show the results of a prototype chip-embedded interposer with a panel size of 515mm x 510mm. The interposer size was 5.9 reticle size, and 20 chips were embedded in one interposer. Fig.5 shows the panel appearance after the process and bridge chip mounting. Cu pillars were formed using Resonac's dry film resist with a thickness of 112 μm. The bridge chips were thermocompression-bonded using die-attach film, achieving accuracy of less than 2.5μm. Alignment between the Cu pillars after grinding and the vias of the RDL wiring was significantly improved by performing die-by-die alignment during exposure. As a result, we successfully fabricated a chip-embedded interposer with a minimum wiring pitch of 10μm at the panel size.

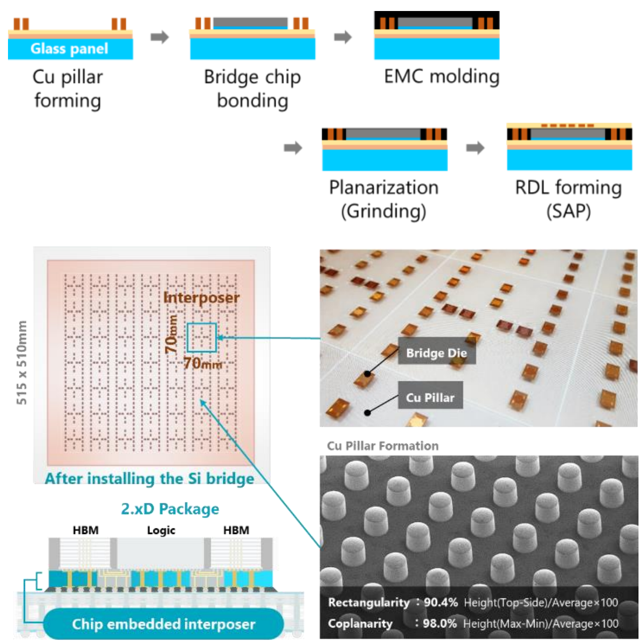


Fig.6 Chip embedded after the process of bridge chip bonding

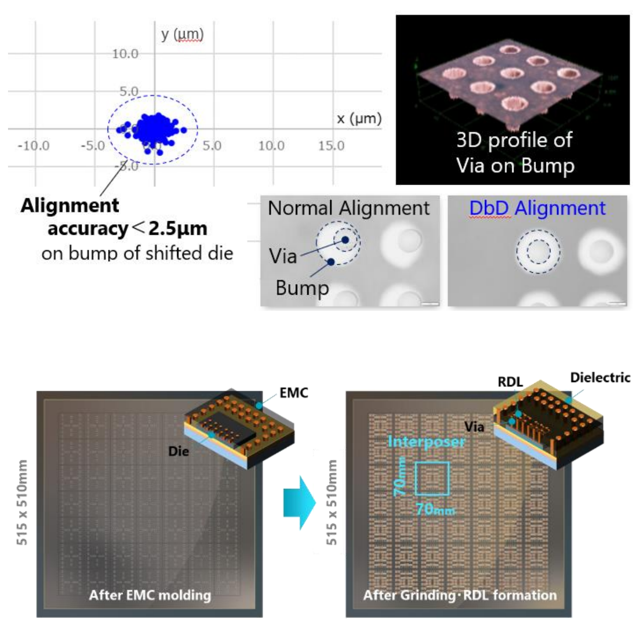
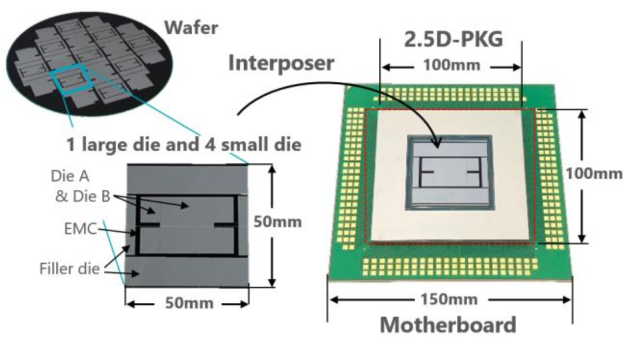


Fig.7 Panel level chip embedded interposer (510 x 515mm size)

6. High reliability large package (WG-C)

Integration of chips using interposers has accelerated the trend toward increasingly large package sizes. As a result, modern substrates are expected to deliver not only high-density interconnections and superior electrical performance, but also effective mitigation of warpage arising from mismatches in the coefficient of thermal expansion (CTE) among the chip, interposer, and substrate. With the continued scaling of package dimensions, warpage induced by the CTE gap between the interposer and substrate has become more pronounced, making its minimization a critical requirement for ensuring package-level reliability. Addressing this issue demands a comprehensive approach that combines design optimization with process innovation. In the work of WG-C, we aim to experimentally assess the impact of warpage in ultra-large advanced packages. To this end, we designed and fabricated two 2.5D advanced packages with dimensions of 100 mm × 100 mm and 120 mm × 120 mm, respectively. And then these packages were mounted on a 150 x 150 mm² motherboard. Reliability evaluations were conducted through temperature cycling tests, enabling a systematic investigation of how the CTE of the substrate core material and the solder joint structure influence both thermal reliability and warpage behavior.

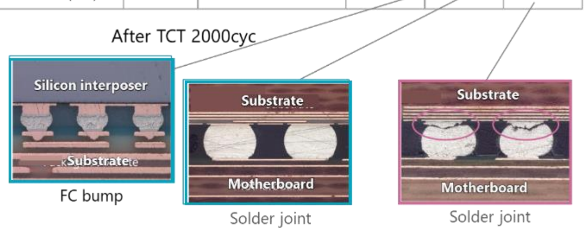


Substrate structure/specification

Size		**100x100mm**
Structure (Layer)		4/2/4
Total thickness		1.6mm
Core	Item	MCL-E-705G(LH)
	Thickness	1.3mm
ABF	Film thickness	15µm
SR	Thickness (on Copper)	18µm

Fig.8 Specification of 100 x 100mm size 2.5D-package

Substrate		Motherboard		TCT 1000cyc	TCT 2000cyc
Core material	CTE1 [ppm/°C]	Core material	CTE1 [ppm/°C]		
E-705G(LH)	6.9	E-705G(LH)	6.9	Pass	Pass
E-705G(LH)	6.9	Conventional	15.0	Pass	Fail



*The criteria was set at resistance change of 20% or less.

Fig.9 TCT (-55ºC/125ºC) after mounting on motherboard

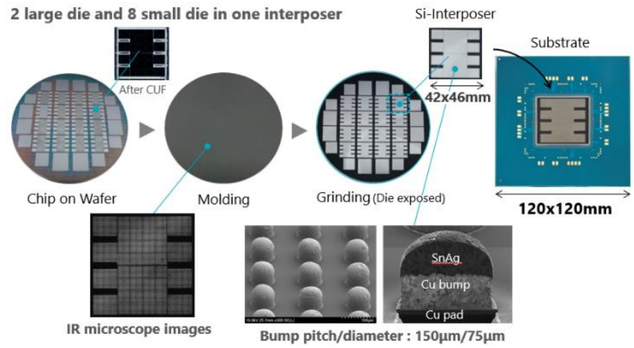


Fig.10 Interposer process for 120 x 120mm size 2.5D-package

7. Conclusion

Through the JOINT2 evaluation platform, we verified key technologies for next-generation packaging. WG-A demonstrated 10 µm-pitch interconnections using the CoW process and identified material properties essential for miniaturization. WG-B fabricated multi-layer RDL and die-embedded interposers at the panel level, and evaluated the applicability of a PID-based damascene process for L/S = 1/1 µm wiring. WG-C developed a 120 mm-square 2.5D package that achieved 1,000 cycles in reliability testing. These achievements provide valuable guidelines for advancing equipment, process technologies, and material properties. As next steps, ensuring the reliability of large packages with panel-level interposers and optimizing overall structural systems—including thermal management and warpage control—are priorities. To address these challenges, we will accelerate the provision of suitable materials and processes by strengthening collaboration across design, equipment, and materials, and by creating new opportunities for co-creation, thereby promoting technology dissemination and innovation across the industry.

8. Acknowledgement

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9. References

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